

U-Net: Convolutional Networks for Biomedical Image Segmentation

Olaf Ronneberger, Philipp Fischer, and Thomas Brox

Computer Science Department and BIOSS Centre for Biological Signalling Studies,
University of Freiburg, Germany
`ronneber@informatik.uni-freiburg.de`,
WWW home page: <http://lmb.informatik.uni-freiburg.de/>

Abstract. There is large consent that successful training of deep networks requires many thousand annotated training samples. In this paper, we present a network and training strategy that relies on the strong use of data augmentation to use the available annotated samples more efficiently. The architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. We show that such a network can be trained end-to-end from very few images and outperforms the prior best method (a sliding-window convolutional network) on the ISBI challenge for segmentation of neuronal structures in electron microscopic stacks. Using the same network trained on transmitted light microscopy images (phase contrast and DIC) we won the ISBI cell tracking challenge 2015 in these categories by a large margin. Moreover, the network is fast. Segmentation of a 512x512 image takes less than a second on a recent GPU. The full implementation (based on Caffe) and the trained networks are available at <http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net>.

1 Introduction

In the last two years, deep convolutional networks have outperformed the state of the art in many visual recognition tasks, e.g. [7,3]. While convolutional networks have already existed for a long time [8], their success was limited due to the size of the available training sets and the size of the considered networks. The breakthrough by Krizhevsky et al. [7] was due to supervised training of a large network with 8 layers and millions of parameters on the ImageNet dataset with 1 million training images. Since then, even larger and deeper networks have been trained [12].

The typical use of convolutional networks is on classification tasks, where the output to an image is a single class label. However, in many visual tasks, especially in biomedical image processing, the desired output should include localization, i.e., a class label is supposed to be assigned to each pixel. Moreover, thousands of training images are usually beyond reach in biomedical tasks. Hence, Ciresan et al. [1] trained a network in a sliding-window setup to predict the class label of each pixel by providing a local region (patch) around that pixel

U-Net: 用于生物医学图像分割的卷积网络

Olaf Ronneberger、Philipp Fischer 和 Thomas Brox

德国弗莱堡大学计算机科学系与BIOSS生物信号研究中心
`ronneber@informatik.uni-freiburg.de`, WWW主页:
<http://lmb.informatik.uni-freiburg.de/>

摘要。人们普遍认为，深度网络的成功训练需要数千个标注训练样本。在本文中，我们提出了一种网络和训练策略，该策略依赖于大量使用**数据增强**，以更高效地利用可用的标注样本。该架构由一个用于**捕获上下文的收缩路径**和一个**能够实现精确定位的对称扩展**路径组成。我们表明，这样的网络可以从极少的图像中进行端到端训练，并且在ISBI挑战赛中**对电子显微镜堆栈中神经元结构的分割任务上**，优于此前的最佳方法（滑动窗口卷积网络）。使用在透射光显微镜图像（相差和DIC）上训练的同一网络，我们在这些类别中以较大优势赢得了2015年ISBI细胞追踪挑战赛。此外，该网络速度很快。在最近的GPU上，分割一张512x512图像耗时不到一秒。完整实现（基于Caffe）和训练好的网络可在 <http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net> 获取。

1 引言

在过去两年中，深度卷积网络在许多视觉识别任务上超越了当前最优方法，例如 [7,3]。虽然卷积网络早已存在 [8]，但由于可用训练集的规模以及所考虑网络的规模有限，其成功受到了限制。Krizhevsky等人的突破 [7] 归功于在包含100万张训练图像的ImageNet数据集上对具有8层和数百万参数的大型网络进行监督训练。此后，人们训练了更大、更深的网络 [12]。

卷积网络的典型应用是分类任务，其中对图像的输出是一个单一的类别标签。然而，在许多视觉任务中，尤其是在生物医学图像处理中，期望的输出应包含定位，即**每个像素都应被分配一个类别标签**。此外，在生物医学任务中，通常无法获取数千张训练图像。因此，Ciresan等人 [1] 在滑动窗口设置中训练了一个网络，通过提供该像素周围的局部区域（图像块）作为输入，来预测每个像素的类别标签

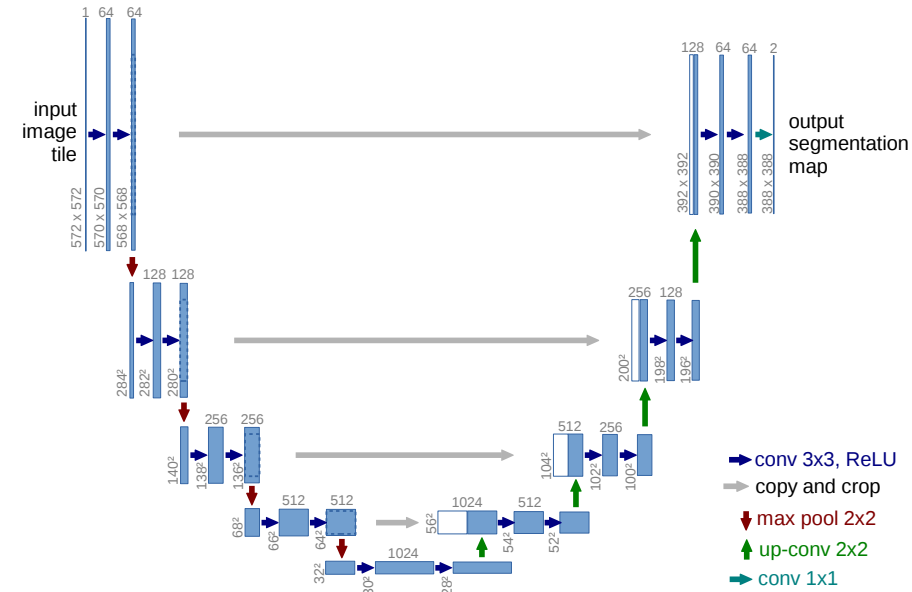


Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

as input. First, this network can localize. Secondly, the training data in terms of patches is much larger than the number of training images. The resulting network won the EM segmentation challenge at ISBI 2012 by a large margin.

Obviously, the strategy in Ciresan et al. [1] has two drawbacks. First, it is quite slow because the network must be run separately for each patch, and there is a lot of redundancy due to overlapping patches. Secondly, there is a trade-off between localization accuracy and the use of context. Larger patches require more max-pooling layers that reduce the localization accuracy, while small patches allow the network to see only little context. More recent approaches [11,4] proposed a classifier output that takes into account the features from multiple layers. Good localization and the use of context are possible at the same time.

In this paper, we build upon a more elegant architecture, the so-called “fully convolutional network” [9]. We modify and extend this architecture such that it works with very few training images and yields more precise segmentations; see Figure 1. The main idea in [9] is to supplement a usual contracting network by successive layers, where pooling operators are replaced by upsampling operators. Hence, these layers increase the resolution of the output. In order to localize, high resolution features from the contracting path are combined with the upsampled

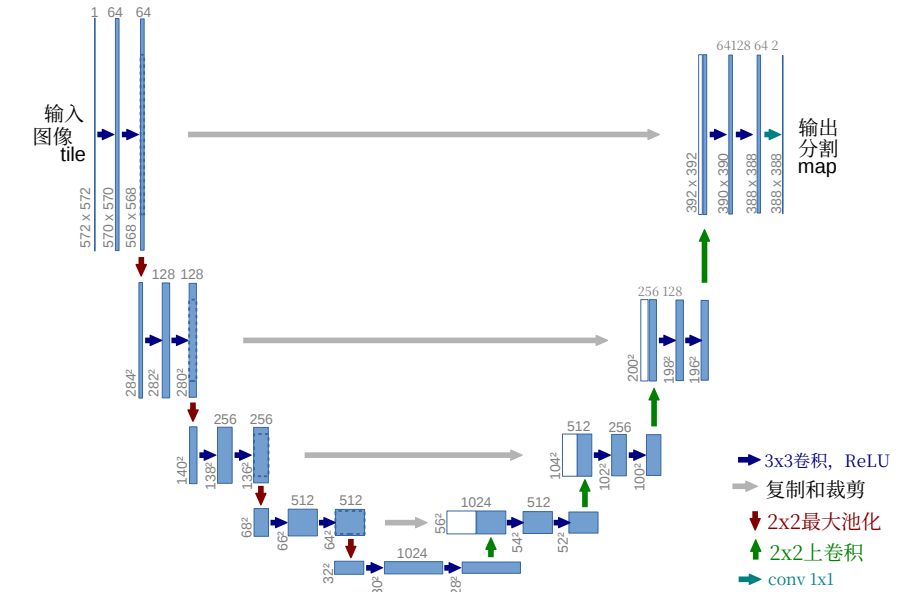


图 1. U-Net 架构（以最低分辨率为 32x32 像素为例）。每个蓝色方框对应一个多通道特征图。通道数标注在方框顶部。x-y尺寸标注在方框左下边缘。白色方框表示复制的特征图。箭头表示不同的操作。

。首先，该网络能够进行定位。其次，以图像块形式呈现的训练数据远多于训练图像的数量。最终得到的网络以较大优势赢得了ISBI 2012的EM分割挑战赛。

显然，Ciresan等人 [1] 的策略有两个缺点。首先，它速度相当慢，因为必须为每个图像块单独运行网络，而且由于图像块重叠，存在大量冗余。其次，在定位精度与上下文利用之间存在权衡。较大的图像块需要更多的最大池化层，这会降低定位精度，而较小的图像块则使网络只能看到很少的上下文。更近期的方法[11,4]提出了一种考虑多层特征的分类器输出。这样就能同时实现良好的定位和对上下文的利用。

在本文中，我们基于一种更优雅的架构，即所谓的“全卷积网络” [9]。我们修改并扩展了该架构，使其能够在极少量训练图像的情况下工作，并产生更精确的分割结果；参见图1。其核心思想在于 [9] 通过逐层叠加来补充通常的收缩网络，其中池化算子被上采样算子所替代。因此，这些层提高了输出的分辨率。为了实现定位，来自收缩路径的高分辨率特征与上采样后的

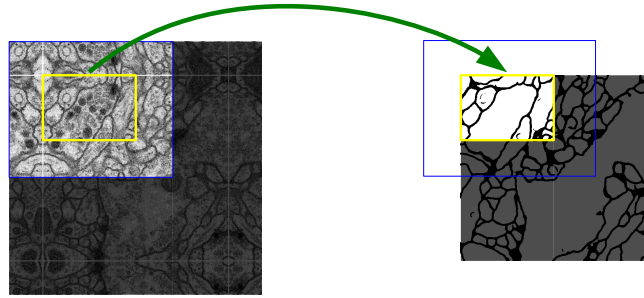


Fig. 2. Overlap-tile strategy for seamless segmentation of arbitrary large images (here segmentation of neuronal structures in EM stacks). Prediction of the segmentation in the yellow area, requires image data within the blue area as input. Missing input data is extrapolated by mirroring

output. A successive convolution layer can then learn to assemble a more precise output based on this information.

One important modification in our architecture is that in the upsampling part we have also a large number of feature channels, which allow the network to propagate context information to higher resolution layers. As a consequence, the expansive path is more or less symmetric to the contracting path, and yields a u-shaped architecture. The network does not have any fully connected layers and only uses the valid part of each convolution, i.e., the segmentation map only contains the pixels, for which the full context is available in the input image. This strategy allows the seamless segmentation of arbitrarily large images by an overlap-tile strategy (see Figure 2). To predict the pixels in the border region of the image, the missing context is extrapolated by mirroring the input image. This tiling strategy is important to apply the network to large images, since otherwise the resolution would be limited by the GPU memory.

As for our tasks there is very little training data available, we use excessive data augmentation by applying elastic deformations to the available training images. This allows the network to learn invariance to such deformations, without the need to see these transformations in the annotated image corpus. This is particularly important in biomedical segmentation, since deformation used to be the most common variation in tissue and realistic deformations can be simulated efficiently. The value of data augmentation for learning invariance has been shown in Dosovitskiy et al. [2] in the scope of unsupervised feature learning.

Another challenge in many cell segmentation tasks is the separation of touching objects of the same class; see Figure 3. To this end, we propose the use of a weighted loss, where the separating background labels between touching cells obtain a large weight in the loss function.

The resulting network is applicable to various biomedical segmentation problems. In this paper, we show results on the segmentation of neuronal structures in EM stacks (an ongoing competition started at ISBI 2012), where we out-

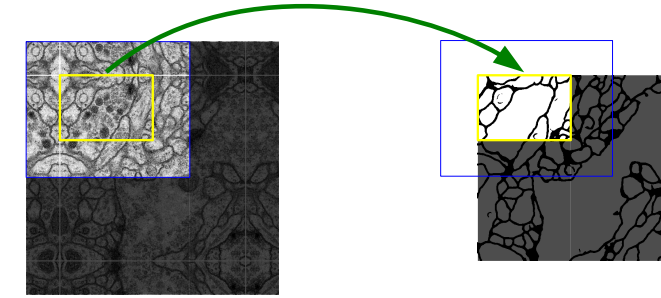


图2. 用于任意大图像无缝分割的重叠切片策略（此处为EM栈中神经元结构的分割）。预测黄色区域的分割需要蓝色区域内的图像数据作为输入。缺失的输入数据通过镜像进行外推

输出相结合。随后，一个连续的卷积层便能够基于这些信息学习组合出更精确的输出。

我们架构中的一个重要修改是，在上采样部分我们也拥有大量的特征通道，这使得网络能够将上下文信息传播到更高分辨率层。因此，扩展路径与收缩路径大致对称，形成U形架构。该网络没有任何全连接层，仅使用每次卷积的有效部分，即分割图仅包含那些在输入图像中具有完整上下文的像素。该策略允许通过重叠切片策略对任意大的图像进行无缝分割（参见图2）。为了预测图像边界区域中的像素，缺失的上下文通过对输入图像进行镜像来外推。这种切片策略对于将网络应用于大图像非常重要，否则分辨率将受限于GPU内存。

由于我们的任务中可用的训练数据非常少，我们通过对可用的训练图像施加弹性形变来进行过度的数据增强。这使得网络能够学习对这些形变的不变性，而无需在标注图像语料库中看到这些变换。这在生物医学分割中尤为重要，因为形变曾是组织中最常见的变异，而真实的形变可以被高效地模拟。数据增强对学习不变性的价值已在Dosovitskiy等人的工作中得到展示 [2]，其研究范围属于无监督特征学习。

许多细胞分割任务中的另一个挑战是分离同一类别的接触对象；参见图3。为此，我们提出使用加权损失，其中接触细胞之间的分离背景标签在损失函数中获得较大的权重。

由此生成的网络可应用于多种生物医学分割问题。在本文中，我们展示了在EM栈中神经元结构分割的结果（一项始于ISBI 2012的持续竞赛），在该任务中我们的表现

performed the network of Ciresan et al. [1]. Furthermore, we show results for cell segmentation in light microscopy images from the ISBI cell tracking challenge 2015. Here we won with a large margin on the two most challenging 2D transmitted light datasets.

2 Network Architecture

The network architecture is illustrated in Figure 1. It consists of a contracting path (left side) and an expansive path (right side). The contracting path follows the typical architecture of a convolutional network. It consists of the repeated application of two 3x3 convolutions (unpadded convolutions), each followed by a rectified linear unit (ReLU) and a 2x2 max pooling operation with stride 2 for downsampling. At each downsampling step we double the number of feature channels. Every step in the expansive path consists of an upsampling of the feature map followed by a 2x2 convolution (“up-convolution”) that halves the number of feature channels, a concatenation with the correspondingly cropped feature map from the contracting path, and two 3x3 convolutions, each followed by a ReLU. The cropping is necessary due to the loss of border pixels in every convolution. At the final layer a 1x1 convolution is used to map each 64-component feature vector to the desired number of classes. In total the network has 23 convolutional layers.

To allow a seamless tiling of the output segmentation map (see Figure 2), it is important to select the input tile size such that all 2x2 max-pooling operations are applied to a layer with an even x- and y-size.

3 Training

The input images and their corresponding segmentation maps are used to train the network with the stochastic gradient descent implementation of Caffe [6]. Due to the unpadded convolutions, the output image is smaller than the input by a constant border width. To minimize the overhead and make maximum use of the GPU memory, we favor large input tiles over a large batch size and hence reduce the batch to a single image. Accordingly we use a high momentum (0.99) such that a large number of the previously seen training samples determine the update in the current optimization step.

The energy function is computed by a pixel-wise soft-max over the final feature map combined with the cross entropy loss function. The soft-max is defined as $p_k(\mathbf{x}) = \exp(a_k(\mathbf{x})) / \left(\sum_{k'=1}^K \exp(a_{k'}(\mathbf{x})) \right)$ where $a_k(\mathbf{x})$ denotes the activation in feature channel k at the pixel position $\mathbf{x} \in \Omega$ with $\Omega \subset \mathbb{Z}^2$. K is the number of classes and $p_k(\mathbf{x})$ is the approximated maximum-function. I.e. $p_k(\mathbf{x}) \approx 1$ for the k that has the maximum activation $a_k(\mathbf{x})$ and $p_k(\mathbf{x}) \approx 0$ for all other k . The cross entropy then penalizes at each position the deviation of $p_{\ell(\mathbf{x})}(\mathbf{x})$ from 1 using

$$E = \sum_{\mathbf{x} \in \Omega} w(\mathbf{x}) \log(p_{\ell(\mathbf{x})}(\mathbf{x})) \quad (1)$$

超越了Ciresan等人的网络 [1]。此外，我们展示了ISBI细胞追踪挑战赛2015中光学显微镜图像细胞分割的结果。在此，我们在两个最具挑战性的2D透射光数据集上以大幅优势获胜。

2 网络架构

该网络架构如图1所示。图1它由一条收缩路径（左侧）和一条扩展路径（右侧）组成。收缩路径遵循卷积网络的典型架构，由重复应用两个 3x3 卷积（无填充卷积）构成，每个卷积后均接一个修正线性单元（ReLU）和一个步幅为 2 的 2x2 最大池化操作以进行下采样。在每一次下采样步骤中，我们将特征通道的数量加倍。扩展路径中的每一步都包括：对特征图进行上采样，随后进行一个将特征通道数量减半的 2x2 卷积（“上卷积”），与来自收缩路径的相应裁剪特征图进行拼接，以及两个 3x3 卷积，每个卷积后均接一个 ReLU。由于每次卷积都会丢失边界像素，因此裁剪是必要的。在最后一层，使用一个 1x1 卷积将每个 64 维特征向量映射到所需数量的类别。总体而言，该网络包含 23 个卷积层。

为了实现输出分割图的无缝拼接（见图2），选择输入瓦片尺寸时，必须确保所有 2×2 最大池化操作都应用于x和y尺寸均为偶数的层。

3 训练

使用输入图像及其对应的分割图，通过Caffe [6]的随机梯度下降实现来训练网络。由于采用无填充卷积，输出图像比输入图像小一个固定的边界宽度。为了最小化开销并最大限度地利用GPU内存，我们倾向于使用较大的输入切片而非较大的批量大小，因此将批量减小为单张图像。相应地，我们使用较高的动量（0.99），使得大量先前见过的训练样本决定当前优化步骤中的更新。

能量函数由最终特征图上的逐像素soft-max与交叉熵损失函数共同计算得到。soft-max定义为 $p_k(\mathbf{x}) = \exp(a_k(\mathbf{x})) / \left(\sum_{k'=1}^K \exp(a_{k'}(\mathbf{x})) \right)$ 其中 $a_k(\mathbf{x})$ 表示特征通道 k 中像素位置 $\mathbf{x} \in \Omega$ 处的激活，且 $\Omega \subset \mathbb{Z}^2$ 。 K 是类别数量， $p_k(\mathbf{x})$ 是近似的最大值函数。即， $p_k(\mathbf{x}) \approx 1$ 对于具有最大激活 $a_k(\mathbf{x})$ 的 k ，而 $p_k(\mathbf{x}) \approx 0$ 对于所有其他 k 。交叉熵随后在每个位置惩罚 $p_{\ell(\mathbf{x})}(\mathbf{x})$ 与1的偏差，使用

$$E = \sum_{\mathbf{x} \in \Omega} w(\mathbf{x}) \log(p_{\ell(\mathbf{x})}(\mathbf{x})) \quad (1)$$

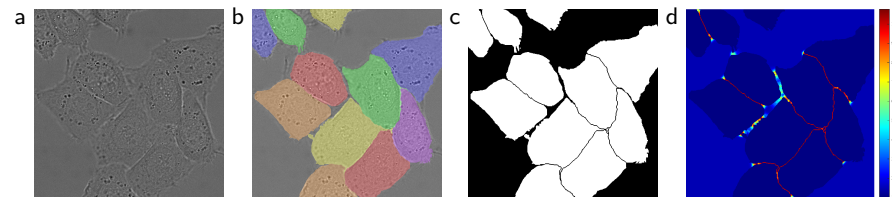


Fig. 3. HeLa cells on glass recorded with DIC (differential interference contrast) microscopy. (a) raw image. (b) overlay with ground truth segmentation. Different colors indicate different instances of the HeLa cells. (c) generated segmentation mask (white: foreground, black: background). (d) map with a pixel-wise loss weight to force the network to learn the border pixels.

where $\ell : \Omega \rightarrow \{1, \dots, K\}$ is the true label of each pixel and $w : \Omega \rightarrow \mathbb{R}$ is a weight map that we introduced to give some pixels more importance in the training.

We pre-compute the weight map for each ground truth segmentation to compensate the different frequency of pixels from a certain class in the training data set, and to force the network to learn the small separation borders that we introduce between touching cells (See Figure 3c and d).

The separation border is computed using morphological operations. The weight map is then computed as

$$w(\mathbf{x}) = w_c(\mathbf{x}) + w_0 \cdot \exp\left(-\frac{(d_1(\mathbf{x}) + d_2(\mathbf{x}))^2}{2\sigma^2}\right) \quad (2)$$

where $w_c : \Omega \rightarrow \mathbb{R}$ is the weight map to balance the class frequencies, $d_1 : \Omega \rightarrow \mathbb{R}$ denotes the distance to the border of the nearest cell and $d_2 : \Omega \rightarrow \mathbb{R}$ the distance to the border of the second nearest cell. In our experiments we set $w_0 = 10$ and $\sigma \approx 5$ pixels.

In deep networks with many convolutional layers and different paths through the network, a good initialization of the weights is extremely important. Otherwise, parts of the network might give excessive activations, while other parts never contribute. Ideally the initial weights should be adapted such that each feature map in the network has approximately unit variance. For a network with our architecture (alternating convolution and ReLU layers) this can be achieved by drawing the initial weights from a Gaussian distribution with a standard deviation of $\sqrt{2/N}$, where N denotes the number of incoming nodes of one neuron [5]. E.g. for a 3x3 convolution and 64 feature channels in the previous layer $N = 9 \cdot 64 = 576$.

3.1 Data Augmentation

Data augmentation is essential to teach the network the desired invariance and robustness properties, when only few training samples are available. In case of

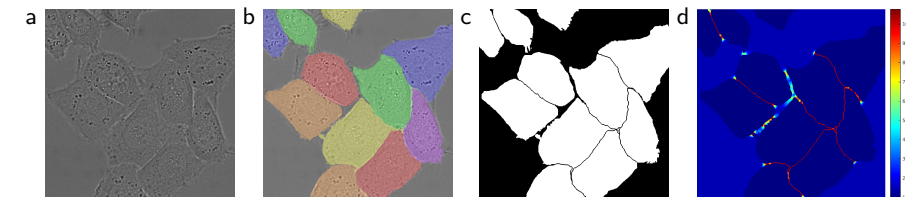


图3. DIC（微分干涉相差）显微镜记录的玻璃上的HeLa细胞。(a) 原始图像。(b) 与真值分割的叠加。不同颜色表示HeLa细胞的不同实例。(c) 生成的分割掩膜（白色：前景，黑色：背景）。(d) 带有逐像素损失权重的图，用于强制网络学习边界像素。

其中 $\ell : \Omega \rightarrow \{1, \dots, K\}$ 是每个像素的真实标签， $w : \Omega \rightarrow \mathbb{R}$ 是我们在训练中引入的权重图，以赋予某些像素更高的重要性。

我们为每个真值分割预先计算权重图，以补偿训练数据集中来自某一类别的像素的不同频率，并强制网络学习我们在接触细胞之间引入的小分离边界（参见图3c和d）。

分离边界使用形态学操作计算。然后按以下方式计算权重图

$$w(\mathbf{x}) = w_c(\mathbf{x}) + w_0 \cdot \exp\left(-\frac{(d_1(\mathbf{x}) + d_2(\mathbf{x}))^2}{2\sigma^2}\right) \quad (2)$$

其中 $w_c : \Omega \rightarrow \mathbb{R}$ 是用于平衡类别频率的权重图， $d_1 : \Omega \rightarrow \mathbb{R}$ 表示到最近细胞边界的距离， $d_2 : \Omega \rightarrow \mathbb{R}$ 表示到第二近细胞边界的距离。在我们的实验中，我们设置 $w_0 = 10$ 和 $\sigma \approx 5$ 像素。

在具有许多卷积层和网络中不同路径的深度网络中，良好的权重初始化极为重要。否则，网络的一部分可能会产生过大的激活值，而其他部分则从不贡献。理想情况下，初始权重应被调整为使网络中的每个特征图具有近似单位方差。对于采用我们架构的网络（交替的卷积层和ReLU层），这可以通过从标准差为 $\sqrt{2/N}$ 的高斯分布中抽取初始权重来实现，其中 N 表示一个神经元的输入节点数 [5]。例如，对于3x3卷积且前一层有64个特征通道的情况 $N = 9 \cdot 64 = 576$ 。

3.1 数据增强

数据增强对于在仅有少量训练样本时教会网络所需的不变性和鲁棒性至关重要。在

microscopical images we primarily need shift and rotation invariance as well as robustness to deformations and gray value variations. Especially random elastic deformations of the training samples seem to be the key concept to train a segmentation network with very few annotated images. We generate smooth deformations using random displacement vectors on a coarse 3 by 3 grid. The displacements are sampled from a Gaussian distribution with 10 pixels standard deviation. Per-pixel displacements are then computed using bicubic interpolation. Drop-out layers at the end of the contracting path perform further implicit data augmentation.

4 Experiments

We demonstrate the application of the u-net to three different segmentation tasks. The first task is the segmentation of neuronal structures in electron microscopic recordings. An example of the data set and our obtained segmentation is displayed in Figure 2. We provide the full result as Supplementary Material. The data set is provided by the EM segmentation challenge [14] that was started at ISBI 2012 and is still open for new contributions. The training data is a set of 30 images (512x512 pixels) from serial section transmission electron microscopy of the Drosophila first instar larva ventral nerve cord (VNC). Each image comes with a corresponding fully annotated ground truth segmentation map for cells (white) and membranes (black). The test set is publicly available, but its segmentation maps are kept secret. An evaluation can be obtained by sending the predicted membrane probability map to the organizers. The evaluation is done by thresholding the map at 10 different levels and computation of the “warping error”, the “Rand error” and the “pixel error” [14].

The u-net (averaged over 7 rotated versions of the input data) achieves without any further pre- or postprocessing a warping error of 0.0003529 (the new best score, see Table 1) and a rand-error of 0.0382.

This is significantly better than the sliding-window convolutional network result by Ciresan et al. [1], whose best submission had a warping error of 0.000420 and a rand error of 0.0504. In terms of rand error the only better performing

Table 1. Ranking on the EM segmentation challenge [14] (march 6th, 2015), sorted by warping error.

Rank	Group name	Warping Error	Rand Error	Pixel Error
	** human values **	0.000005	0.0021	0.0010
1.	u-net	0.000353	0.0382	0.0611
2.	DIVE-SCI	0.000355	0.0305	0.0584
3.	IDSIA [1]	0.000420	0.0504	0.0613
4.	DIVE	0.000430	0.0545	0.0582
⋮				
10.	IDSIA-SCI	0.000653	0.0189	0.1027

显微图像的情况下，我们主要需要**平移和旋转不变性，以及对形变和灰度变化**的鲁棒性。特别是训练样本的随机弹性形变似乎是训练具有极少标注图像的分割网络的关键概念。我们在粗 3×3 网格上使用随机位移向量生成平滑形变。位移从标准差为10像素的高斯分布中采样。然后使用双三次插值计算逐像素位移。收缩路径末端的Drop-out层执行进一步的隐式数据增强。

4 实验

我们展示了u-net在三种不同分割任务中的应用。第一个任务是对电子显微镜记录中的神经元结构进行分割。该数据集的一个示例以及我们获得的分割结果展示在图2中。我们在补充材料中提供了完整结果。该数据集由EM分割挑战赛提供 [14]，该挑战赛始于ISBI 2012，目前仍开放接受新的贡献。训练数据是一组30张图像（512x512像素），来自果蝇一龄幼虫腹神经索（VNC）的连续切片透射电子显微镜。每张图像都附带相应的完全标注的细胞（白色）和细胞膜（黑色）真值分割图。测试集是公开可用的，但其分割图保密。可以通过将预测的细胞膜概率图发送给组织者来获得评估。评估通过在10个不同阈值水平上对图进行阈值化，并计算“变形误差”、“Rand误差”和“像素误差”来完成 [14]。

U-Net（对输入数据的7个旋转版本取平均）在没有任何进一步的预处理或后处理的情况下，实现了0.0003529的变形误差（新的最佳分数，见表1）和0.0382的Rand误差。

这显著优于Ciresan等人 [1]，的滑动窗口卷积网络结果，其最佳提交的变形误差为0.000420，Rand误差为0.0504。就Rand误差而言，唯一表现更好的

表 1. EM分割挑战赛排名 [14]（2015年3月6日），按变形误差排序。

Rank	组名	变形误差	Rand误差	像素误差
	** 人工结果 **	0.000005	0.0021	0.0010
1.	U-Net	0.000353	0.0382	0.0611
2.	DIVE-SCI	0.000355	0.0305	0.0584
3.	IDSIA [1]	0.000420	0.0504	0.0613
4.	DIVE	0.000430	0.0545	0.0582
⋮				
10.	IDSIA-SCI	0.000653	0.0189	0.1027

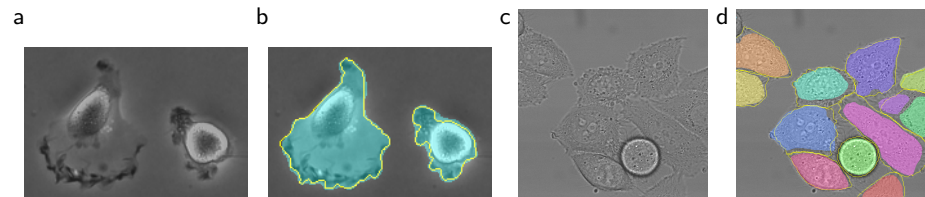


Fig. 4. Result on the ISBI cell tracking challenge. (a) part of an input image of the “PhC-U373” data set. (b) Segmentation result (cyan mask) with manual ground truth (yellow border) (c) input image of the “DIC-HeLa” data set. (d) Segmentation result (random colored masks) with manual ground truth (yellow border).

Table 2. Segmentation results (IOU) on the ISBI cell tracking challenge 2015.

Name	PhC-U373	DIC-HeLa
IMCB-SG (2014)	0.2669	0.2935
KTH-SE (2014)	0.7953	0.4607
HOUS-US (2014)	0.5323	-
second-best 2015	0.83	0.46
u-net (2015)	0.9203	0.7756

algorithms on this data set use highly data set specific post-processing methods¹ applied to the probability map of Cirestan et al. [1].

We also applied the u-net to a cell segmentation task in light microscopic images. This segmenation task is part of the ISBI cell tracking challenge 2014 and 2015 [10,13]. The first data set “PhC-U373”² contains Glioblastoma-astrocytoma U373 cells on a polyacrylimide substrate recorded by phase contrast microscopy (see Figure 4a,b and Supp. Material). It contains 35 partially annotated training images. Here we achieve an average IOU (“intersection over union”) of 92%, which is significantly better than the second best algorithm with 83% (see Table 2). The second data set “DIC-HeLa”³ are HeLa cells on a flat glass recorded by differential interference contrast (DIC) microscopy (see Figure 3, Figure 4c,d and Supp. Material). It contains 20 partially annotated training images. Here we achieve an average IOU of 77.5% which is significantly better than the second best algorithm with 46%.

5 Conclusion

The u-net architecture achieves very good performance on very different biomedical segmentation applications. Thanks to data augmentation with elastic defor-

¹ The authors of this algorithm have submitted 78 different solutions to achieve this result.

² Data set provided by Dr. Sanjay Kumar. Department of Bioengineering University of California at Berkeley. Berkeley CA (USA)

³ Data set provided by Dr. Gert van Cappellen Erasmus Medical Center. Rotterdam. The Netherlands

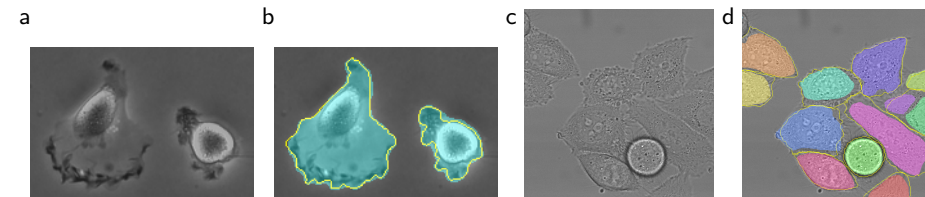


图 4. ISBI细胞追踪挑战赛上的结果。(a) “PhC-U373”数据集输入图像的一部分。(b) 分割结果（青色掩膜）与人工标注真值（黄色边框）(c) “DIC-HeLa”数据集的输入图像。(d) 分割结果（随机彩色掩膜）与人工标注真值（黄色边框）。

表 2. 2015年ISBI细胞追踪挑战赛上的分割结果（交并比）。

Name	PhC-U373	DIC-HeLa
IMCB-SG (2014)	0.2669	0.2935
KTH-SE (2014)	0.7953	0.4607
HOUS-US (2014)	0.5323	-
2015年第二名	0.83	0.46
U-Net (2015)	0.9203	0.7756

该数据集上的算法使用了高度针对该数据集的后处理方法¹应用于Ciresan等人的概率图[1]。

我们还将U-Net应用于光学显微图像中的细胞分割任务。该分割任务是ISBI细胞追踪挑战赛2014的一部分，2015 [10,13]。第一个数据集 “PhC-U373”² 包含在聚丙烯酰胺基底上通过相差显微镜记录的胶质母细胞瘤-星形细胞瘤U373细胞（参见图4a、b及补充材料）。该数据集包含35张部分标注的训练图像。在此我们达到了92%的平均交并比（“交并比”），显著优于第二优算法83%的结果（参见表2）。第二个数据集 “DIC-HeLa”³ 是通过微分干涉相差（DIC）显微镜记录的平坦玻璃上的HeLa细胞（参见图3、图4c、d及补充材料）。该数据集包含20张部分标注的训练图像。在此我们达到了77.5%的平均交并比，显著优于第二优算法46%的结果。

5 结论

U-Net 架构在非常不同的生物医学分割应用中取得了非常好的性能。得益于带有弹性变形的数据增强，

¹ 该算法的作者提交了78种不同的解决方案以取得这一结果。² 数据集由美国加州伯克利分校生物工程系的Sanjay Kumar博士提供。加利福尼亚州伯克利（美国）³ 数据集由荷兰鹿特丹伊拉斯姆斯医学中心的Gert van Cappellen博士提供

mations, it only needs very few annotated images and has a very reasonable training time of only 10 hours on a NVidia Titan GPU (6 GB). We provide the full Caffe[6]-based implementation and the trained networks⁴. We are sure that the u-net architecture can be applied easily to many more tasks.

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⁴ U-net implementation, trained networks and supplementary material available at <http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net>

它只需要非常少量的标注图像，并且训练时间非常合理，在 NVidia TitanGPU（6 GB）上仅需 10 小时。我们提供了完整的基于 Caffe[6]的实现以及训练好的网络⁴。我们确信 U-Net 架构可以轻松应用于更多任务。

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⁴U-Net 实现、训练好的网络及补充材料可在以下网址获取：
<http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net>